

# MAYUR DOKRAS

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## Professional Summary

Software Engineer with 4 years of experience designing, developing and supporting production software across telecom and cybersecurity domains. Experienced in Python, SQL, FastAPI, REST APIs, data modelling, automation and cloud technologies, with hands-on experience building backend services, reporting workflows and data-driven solutions. Strong background in developing scalable Python applications, operational data processing and API integrations, with a growing focus on AI-powered engineering and analytics platforms.

## Skills

- **Core Languages:** Python, Java, SQL, C/C++, JavaScript, Kotlin, Dart, Swift, Go (Learning)
- **Data Engineering & Analytics:** SQL, Snowflake, PostgreSQL, MySQL, Oracle (PL/SQL), Data Modelling, Relational Database Design, ETL & Data Pipelines, Data Warehousing, Database Performance Optimisation, MongoDB, Redis, Firebase
- **Backend & API:** Python, Java, Spring Boot, FastAPI, REST API Design, OpenAPI, Backend Architecture, Microservices, Third-Party API Integration, Authentication & Authorization, OAuth2, API Security, Event-Driven Architecture, Distributed Systems, Asynchronous Processing
- **AI & Agentic Systems:** OpenAI API, LLM Integration, GitHub Copilot, Retrieval-Augmented Generation (RAG), Prompt Engineering, Tool Calling, Structured Outputs, Agentic AI, Workflow Automation, LangChain (Learning), Voice AI Concepts (STT, TTS, VAD, S2S) [Learning]
- **DevOps & Cloud:** AWS (EC2, S3, IAM), GCP, Linux, Git, GitHub, Jenkins, CI/CD, Continuous Delivery, REST API Testing, Production Troubleshooting, Observability, Docker (Learning), Kubernetes (Learning), Terraform (Learning), Ansible (Learning)
- **Mobile & Frontend:** React.js, TypeScript, Android SDK, Kotlin, Flutter, Dart, SwiftUI, Firebase, MVVM, Coroutines, Dependency Injection, HTML5, CSS3
- **Software Engineering:** System Design, Scalable Architecture, Cloud-Native Architecture, Distributed Systems, Microservices, Design Patterns, Clean Architecture, Secure Software Development, Performance Optimisation, Reliability Engineering, Fault Tolerance, Root Cause Analysis, Unit Testing, Code Reviews, Agile, Full SDLC, Stakeholder Management, Technical Mentoring

## Relevant Work Experience

### Backend Developer – Malvox

*University of Southampton*

*February 2025 - July 2025*

- Led backend development for an AI-powered cybersecurity platform, designing scalable backend services, developing secure API workflows, contributing to backend architecture and collaborating closely with international stakeholders throughout the product lifecycle.
- Worked with Python, FastAPI, SQL, PostgreSQL, REST APIs, OAuth2, React and Git to build backend services, integrate UK Government APIs, design relational data models and develop secure data processing workflows.
- Improved platform reliability by implementing secure backend workflows, transactional processing and structured error handling, enabling scalable production deployments and supporting secure data exchange across integrated government services.

### Associate II - Software Engineer

*Capgemini Engineering*

*November 2021 - November 2024*

- Delivered software solutions for the HPE GPON project, developing and supporting GPON optical network devices while designing Python-based monitoring tools, processing operational and production data, collaborating across engineering teams and supporting production deployments.
- Worked with Python, SQL, C, C++, Linux, Jenkins, Git, GDB and Wireshark to develop automation tools for log analysis, operational data processing, reporting workflows, production debugging and engineering analytics across GPON, CFM and CCM protocols.
- Delivered 7+ production features, resolved 40+ production defects and reduced MTTR by 25%, improving production stability, engineering productivity and the reliability of fibre based communication services.

### Freelance Mobile Application Developer

*Self Employed*

*February 2021 - October 2021*

- Worked directly with clients to gather requirements, define user journeys and deliver mobile applications across e-commerce, grocery, delivery and fitness use cases.

- Developed application features using Flutter, Dart, Firebase and REST APIs, including authentication, offline storage, push notifications and Razorpay and PayPal payment integrations.
- Improved delivery speed by creating reusable application flows and integrating Firebase-based backend, authentication and payment components across projects.

## Projects

### AI Platform (Personal Project)

*(In Progress)*

- Designing an AI platform using Python, FastAPI, PostgreSQL and OpenAI APIs, implementing scalable data ingestion, relational data models, retrieval pipelines, ETL pipelines and Retrieval-Augmented Generation (RAG) workflows.
- Applying AI-assisted workflows across the full SDLC, including requirement analysis, solution design, development, testing, documentation and engineering automation.
- Developing agentic workflows focused on tool calling, structured outputs, retrieval quality, hallucination reduction and future voice capabilities using STT, TTS and S2S concepts.

### Link2Feed

*Dissertation Product*

- Built a social welfare application designed to connect people in need with relevant support services, using Kotlin, Android SDK, Firebase and UK Government APIs.
- Implemented MVVM architecture, Coroutines, dependency injection, structured data layers, API integrations, authentication and Android testing.
- Initially developed the product for Android and later began expanding it into a native iOS application using SwiftUI.

## Education

### MSc in Software Engineering

*University of Southampton, UK – 2024/25*

- Completed coursework and dissertation covering relational databases, Oracle (PL/SQL), MongoDB, data modelling, backend development, secure software engineering and UK Government API integration.

### Bachelor's in Information Technology

*AISSMS IOIT, India – 2018/20*

- Developed a full-stack web application using JavaScript, PHP and SQL, implementing custom ranking logic and database-driven features for topic-based image categorization.

### Diploma in Information Technology

*RIT Polytechnic, India – 2015/17*

- Developed a Java-based steganography system using the LSB algorithm, focusing on secure data hiding and file-level information protection.

## Certifications

- Hands-On Essentials: Data Warehousing Workshop (Snowflake, 2026)
- Hands-On Essentials: Collaboration, Marketplace & Cost Estimation Workshop (Snowflake, 2026)
- Claude Code in Action (Anthropic, 2026)
- Cloud Computing (IBM, 2023)
- Amazon Web Services S3 Bucket (Coursera, 2021)
- Dart Functions Framework (Google Cloud, 2023)
- Agile Software Development (Coursera, 2022)
- Java Object-Oriented Programming (LinkedIn Learning, 2021)